MIL-S-43939(GL)

13 January 1976
SUPERSEDING
(see 6.8)

MILITARY SPECIFICATION

SHIRT, MAN'S, POLYESTER/COTTON, DURABLE PRESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyester/cotton broadcloth durable press shirt.

1.2 Classification. The shirt shall be of one type in the following sizes (see 6.2):

FSC 8405

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Schedule of sizes - inches - (mm)

Neck size	Sleeve length									
13 (330)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	--	--	--	--
13-1/2 (343)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	--
14 (356)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
14-1/2 (368)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
15 (381)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
15-1/2 (394)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
16 (406)	29 (737)	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
16-1/2 (419)	--	30 (762)	31 (787)	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	38 (965)
17 (432)	--	--	--	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	--
17-1/2 (445)	--	--	--	32 (813)	33 (838)	34 (864)	35 (889)	36 (914)	37 (940)	--
18 (457)	--	--	--	--	33 (838)	34 (864)	35 (889)	--	--	--

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label; For Clothing, Equipage, and Tentage, (General Use)

MILITARY

MIL-B-17757	-	Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-C-43479	-	Cloth, Broadcloth, Polyester and Cotton
MIL-T-43548	-	Thread, Polyester, Cotton-Covered
MIL-C-43594	-	Cloth, Interlining, Polyester
MIL-T-43624	-	Thread, Polyester, Spun
MIL-P-55010	-	Plastic Sheet, Polyethylene Terephthalate

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1492	-	Provisions for Evaluating Quality of Men's Shirts

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Manual of the American Association of Textile Chemists and Colorists

(Application for copies of the AATCC Manual should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709.)

Standard Color Card of America, 9th Edition

(Application for copies shall be addressed to the Textile Color Card Association of the United States, Inc., 200 Madison Avenue, New York, NY.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 Material.

3.2.1 Basic material. The basic material shall be polyester/cotton broadcloth, Tan 446, conforming to type I, class 3 of MIL-C-43479.

3.2.2 Interlining.

3.2.2.1 Collarstand. The material for interlining the collarstand shall be polyester fiber cloth conforming to MIL-C-43594.

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3.2.2.2 Collar leaf and cuffs. The material for interlining the collar leaf and cuffs shall be an undyed, preshrunk, 100 percent cotton, plain weave cloth which has a uniformly distributed dot system of high density polyethylene fusible adhesive applied to one side in accordance with good commercial practice. The fusible interlining shall meet the following requirements when tested as specified in 4.2.1.

Weight, oz. per sq. yd.		4.5 ± 0.5
Weave		Plain
Thread per inch (min)	warp	44
	filling	40
Shrinkage, percent (max)	warp	2.0
	filling	2.0

In addition, the fusible interlining shall meet the following requirements after it has been fused to the basic cloth, when tested as specified in 4.2.1.

Stiffness, inch lbs	warp	.010 - .020
	filling	.003 - .008
Adhesion after 25 launderings oz/inch (min)		20

3.2.2.2.1 Fusing conditions. The fabric shall be fused on a fusing press. For optimum results with the particular fusible style and press used, specific settings should be obtained from the supplier of the fusible material. See 6.7 for recommended fusing conditions.

3.2.3 Labels. Each shirt shall have an identification label, size label and instruction label conforming to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.2.3.1 Identification label. The identification label shall conform to type IV, class 1.

3.2.3.2 Size label. The size label shall conform to type IV, class 2.

3.2.3.3 Instruction label. The instruction label shall conform to type I, class 3, except that the label cloth shall be dyed Lemon Yellow, Cable No. 7020, in conformance with the Standard Color Card of America. The label shall contain the following information.

SHIRT, MAN'S, POLYESTER/COTTON, DURABLE PRESS

Washing Instructions

1. For machine washing, use wash and wear cycle. Wash with moderately hot water (120°F), (49°C), using soap or mild detergent.

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2. For hand washing, wash in moderately hot water (120°F), (49°C), using soap or mild detergent. DO NOT WRING OR TWIST.

3. Dry in tumble dryer at warm setting. Remove immediately at end of drying. Shape and hang.

4. To drip dry, remove from water before last spin cycle. Shape and hang.

5. If touch up is required, iron at a moderate heat setting.

DO NOT REMOVE THIS LABEL

3.2.3.4 Combination identification-instruction label. An identification label (type I, class 1) and an instruction label (type I, class 3) may be combined into one label. When combined, the color of the label shall conform to the requirements of the instruction label (3.2.3.3) and the item nomenclature shall not be repeated. The coating requirements of the type I, class 3 label shall apply.

3.2.4 Thread. The thread for seaming, stitching, buttonholes and button sewing shall be cotton-covered polyester, ticket No. 70, 2 or 3 ply and 80, 2 or 3 ply conforming to MIL-T-43548. Spun polyester thread, ticket Nos. 70, 2 or 3 ply and 90, 2 ply, conforming to class 1 or 2 of MIL-T-43624 may be substituted for the cotton-covered polyester thread.

3.2.4.1 Color and colorfastness. The thread shall be Tan X, C.A. 66041. The dyed thread shall show fastness to light, perspiration, and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light, perspiration, and laundering.

3.2.5 Buttons. The buttons shall conform to type II, class K, style 25, 19 line of V-B-871. The color of the buttons shall be shade DOD Tan AJ, Cable No. 62028.

3.2.6 Collar stays. The collar stays shall be made from polyethylene terephthalate plastic sheet conforming to MIL-P-55010, except the thickness shall be 0.010 ± 0.001 inch (.254 ± .025 mm), the color shall be natural (clear or opaque), and the transparency and optical properties shall not apply. The stays shall measure not less than 1/4 inch (6.4 mm) or more than 3/8 inch (9.5 mm) wide and 2-1/4 to 2-5/16 inches (57 to 59 mm) long. Either both ends shall be rounded or pointed with the points rounded, or one end rounded and one end cut at an angle with the point rounded. Testing shall be as specified in 4.2.1.

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3.3 Design. The shirt is a dress type with patch pocket on left front; the sleeves have continuous bindings on openings and one-button cuffs; the collar and cuffs have fusible interlinings; the collars have stays; and the shirt is a safety-stitch construction and has a yoke.

3.4 Figure. Figure 1 is furnished for information purposes only. If there are any inconsistencies between the specification and the figure, the specification shall control.

3.5 Patterns. Standard patterns will be furnished by the Government and, unless otherwise indicated, provide an allowance of 1/4 inch (6.4 mm) for all seams except 3/8 inch (9.5 mm) for the hem and attaching the cuffs, and 1/2 inch (12.7 mm) for setting the pockets. The standard patterns shall not be altered in any way and are to be used only as a guide for cutting the supplier's working patterns, which shall be identical to the standard patterns.

3.5.1 List of pattern parts. The shirts shall be cut from material as specified in accordance with pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Basic material	Front	2
	Back	1
	Yoke	2
	Sleeve	2
	Sleeve facing (binding)	2
	Cuff 1/	4
	Collar leaf	2
	Collarstand	2
	Pocket	1
	Stay holder	2
Polyester fiber interlining cloth	Collarstand interlining	1
Fusible interlining cloth	Cuff interlining	2
	Collar leaf interlining	1

1/ When optional cuff setting operation is used (operation 17.c), 1/8 inch (3.2 mm) may be added to the patterns. However, finished cuff measurements must be maintained.

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3.6 End item seam appearance and bond strength of fusibles. After the shirt has been subjected to 5 washings, the seams of the shirt shall have a minimum average rating of 3.5, when tested as specified in 4.2.3. This requirement shall be applicable to all seams including front, cuffs, pocket and collar. The shirt shall be subjected to an additional 20 washings (25 total) and shall show no bubbling or delamination of the fused plies of material when tested as specified in 4.2.3. There shall be no "strike-through" of the adhesive after fusing.

3.7 Construction.

3.7.1 Stitches, seams and stitching. Stitch, seam and stitching types specified in table II shall conform to FED-STD-751. Where two or more seam or stitch types are given for the same part of an operation, any one of them may be used. Where stitch type 401 is used, the looper (under thread) shall be on the inside of the shirt. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained (see 6.5) so that there is no tight or loose stitching. The gage of the safety stitch seams shall be 1/4 inch (6.4 mm). The guides and knives on safety stitch machines shall be set to trim only the ravelled ends of the fabric. The trimming shall not exceed 1/16 inch (1.6 mm). The edge and top stitching shall be uniformly gaged. Seams required to be worked out, that have a depth between the fold of 1/16 inch (1.6 mm) or more shall be considered a defect.

3.7.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table II.

3.7.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitchings, shall be backtacked not less than 1/4 inch (6.4 mm). Ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch (12.7 mm). The ends of the label stitching shall be overlapped not less than three stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch (12.7 mm). Skipped stitches or thread breaks on 401 stitch type may be repaired using 301 stitch type.

3.7.4 Buttonholes. The buttonholes shall be straight-cut, purl or whip type and shall fit a size 19 line button. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. When a purl-type buttonhole is used, the purling shall be on the outside of the shirt. All loose thread ends shall be trimmed or fused to within 1/8 inch (3.2 mm) of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

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3.8 Manufacturing operations requirements. The shirt shall be made in accordance with operation requirements specified in table II. The supplier is not required to follow exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished shirt.

3.8.1 Marking. The parts of the shirt shall be marked to insure a uniform shade and size throughout the garment. Any method of marking may be used except:

(a) Metal fastening devices.
(b) Sew-on type tickets.
(c) Adhesive type tickets which leave traces of adhesive on the material after removal of the tickets.

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
1.	Cutting. a. Cut shirts in strict accordance with patterns furnished which show directional lines and marks, and notches for proper assembly. Directional lines shall be placed in the warp direction. When fronts of shirts are cut from the uneven edge of the lay in order to utilize selvages, they shall be blocked out and relaid so that the width of the facings shall be exactly as the pattern. b. Cut component parts from the same piece of material except bindings for sleeve openings; collar stands; under plies of cuffs, collar leaf and yoke which may be cut from ends. Parts cut from ends shall approximate the shade of the shirt. The binding for sleeve opening may be roll cut providing that the finished width of the binding conforms to the measurements cited in operation 12.a. c. Cut the fusible and polyester fiber interlinings in accordance with directional lines on patterns.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1492 shall be removed from production and replaced with non-defective and properly matched components.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
3.	Marking (see 3.8.1). Mark, ticket or bundle all parts of the shirt except those parts cut from ends to insure a uniform shade and size throughout the garment.						
4.	Labeling. a. Center size marking on topcollar stand (1/4 inch (6.4 mm) off center tolerance).						
	b. Position the yellow instruction label on the outside right front (as worn) so that on the finished shirt the bottom of the label shall be 1 to 2 inches (25 to 51 mm) from the hemmed bottom and the side of the label shall be 2-1/4 to 3 inches (57 to 76 mm) from the front folded edge and stitch on all four sides. The stitching shall not be through the printing.	301	LSbj-1	8-14	70-2	70-2	
	c. Apply identification label on outside left front (as worn) with the bottom of label 1 to 2 inches (25 to 51 mm) from the hemmed bottom and the side of the label shall be 2-1/4 to 3 inches (57 to 76 mm) from front folded edge.						
	d. The yellow combination label, when applicable, shall be positioned as in "b" above.	301	LSbj-1	8-14	70-2	70-2	
5.	Make cuffs. Finished appearance. The buttonholes shall be clean cut with the stitching securely caught in the fabric. The cuffs shall finish 2-1/2 $\pm$ 1/4 inches (64 $\pm$ 6 mm) wide and have rounded corners.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
5.	Make cuffs. (cont'd)						
	a. Position fusible interlining 1/16 inch (1.6 mm) from bottom and side edges of outer ply and fuse (see 3.2.2.2.1).						
	b. Fold top edge of the outer ply of cuff over and even with the interlining and stitch 1/4 to 5/16 inch (6.35 to 8.0 mm) from folded edge.	301 or 401	SSbc-1 SSbc-1	10-14 10-14	70-2 70-2	70-2 80-2	
	c. Join the plies of the cuff. Turn, force out edges and single stitch 1/16 to 1/8 inch (1.6 to 3.2 mm) from edge. The edge stitching shall finish 1/2 ± 1/8 inch (12.7 ± 3.2 mm) from top edge of finished cuff. The edge stitching may taper to edge of cuff at ends of stitching and need not be back-stitched.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b)	10-14 10-14 10-14	70-2 70-2 70-2	70-2 80-2 70-2	
	d. Make a horizontal buttonhole in center of each cuff (1/4 inch (6.4 mm) off center tolerance per mitted) with the end of the buttonhole 3/8 ± 1/4, - 1/8) inches (9.5 (± 6.4, - 3.2 mm) from front edge.	304	whip type	42-46 per buttonhole including tack	70-2 or	70-2 or	
	e. Press cuffs on a hot head pressing machine (see operation 22).	304	purl type	42-46 per buttonhole including tack	70-2	70-2	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
6.	Make collar.						
	a. Position fusible interlining to top collar 1/16 (1.6 mm) inch from top and side edges and fuse (see 3.2.2.2.1).	301	SSa-1	6-14	70-2	70-2	
	b. Position stay holder to undercollar with side and bottom edges even and stitch along bottom edges to hold stay holder in place.						
	c. Form a tunnel for the collar stays by stitching the stay holder to undercollar with two rows of stitching 3/8 inch (9.5 mm) apart. The tunnels shall be positioned so that on the finished collar they are at a 25 to 30 degree angle from the front edge of the collar. The collar points shall be aligned with the center of the two rows of stitching and the stitching shall extend to the raw edges. Insert collar stays into the tunnels with tapered ends toward collar points.	301	SSv-2	10-14	70-2	70-2	
	d. With the collar and under collar face to face, stitch around side and top edge.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	70-2 70-2	70-2 80-2	
	e. Trim, turn and press collar points on a collar pointing machine.						
	f. Edge stitch collar 1/16 to 1/8 inch (1.6 to 3.2 mm) from edge.	301	SSe-2(b)	10-14	70-2	70-2	

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					NEEDLE	BOBBIN or LOOPER	COVER
7.	Make collarstand and attach collar.						
	a. Fold back the bottom edge of the undercollar stand 3/16 to 1/4 inch (4.8 to 6.4 mm) and stitch 1/16 to 1/8 inch (1.6 to 3.2 mm) from folded edge. This operation may be omitted.	301	EFa-1	10-14	70-2	70-2	
	b. Superimpose interlining and top collarstand and stitch 1/16 to 1/8 inch (1.6 to 3.2 mm) from top edge from end to end. This operation may be omitted.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 80-2	
	c. Place collar between the top and undercollar stand (with attached interlining) as indicated by notches on pattern and stitch from end to end along top edge of collarstand, 3/16 to 1/4 inch (4.8 to 6.4 mm) from edge. Turn top and undercollar stand to finished position and raise stitch 1/16 to 3/32 inch (1.6 to 2.4 mm) from turned edge. The stitching may continue around the ends of the collarstand (see operation 18.d).	301 or 401 and 301	SSq-2(a) SSq-2(a) SSq-2(b)	10-14 10-14 10-14	70-2 70-2 70-2	70-2 80-2 70-2	
	d. Press collar and collarstand with a hot head pressing machine (see operation 22).						
8.	Hem breast pocket.						
	Overedge stitch top raw edge of pocket and turn top of pocket down to form hem. The hem shall finish 1 inch ± 1/8 inch (25.4 mm ± 3.2 mm).	502 or 503	EFd-1 EFd-1	10-14 10-14	80-2 80-2	80-2 80-2	
9.	Make fronts.						
	a. Hem back edge of left and right facing (if not selvage or if selvage is cut or frayed). The hems shall finish 1/8 to 1/4 inch (3.2 to 6.4 mm) wide.	301 or 401	EFb-1 EFb-1	10-14 10-14	70-2 70-2	70-2 80-2	
	- or -						
	b. Overedge back edge of right and left front.	502 or 503	EFd-1 EFd-1	10-14 10-14	80-2 80-2	80-2 80-2	

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					NEEDLE	BOBBIN or LOOPER	COVER
10.	Set pocket.						
	Finished appearance. The pocket shall be parallel to the front edge of shirt. The pocket for all sizes shall finish 4-1/2 ± 1/4 inches (114 ± 6 mm) wide, 5-1/4 ± 1/4 inches (113 ± 6 mm) deep at center, and 4-3/4 ± 1/4 inches (121 ± 6 mm) deep at sides. The side dimensions shall not vary from each other by more than 1/8 inch (3.2 mm) and the difference between the center and side dimensions shall be not less than 1/2 (12.7 mm) or more than 3.4 inch (19.0 mm)						
	a. Turn in raw edges of pocket 1/2 inch (12.7 mm) and stitch to left front, positioned as indicated by pattern, 1/16 to 1/8 inch (1.6 to 3.2 mm) from edge; continue the stitching across top of pocket for a distance of 3/16 to 1/4 inch (4.8 to 6.4 mm) and down to edge of hem, diagonally or in a straight line. The pockets may be precreased.	301	LSd-1	10-14	70-2	70-2	
	b. Press right and left front with a hot head pressing machine (see operation 22).						
11.	Attach yoke.						
	Finished appearance. The finished yoke shall be without twists or puckers. There shall be no fullness between the two plies.						
	a. Position back of shirt between the two plies of yoke and with the back and yoke edges even, stitch all plies together.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 80-2	
	b. Press back and yoke, with yoke extended, with a hot head pressing machine (see operation 22).						

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					NEEDLE	BOBBIN or LOOPER	COVER
12.	Make sleeve openings.						
	Finished appearance. The binding of the sleeve opening shall be uniform in width and shall finish smooth and flat. Sleeve opening shall measure 5-1/2 ± 1/2 inches (140 ± 13 mm) long exclusive of cuff.						
	a. Turn raw edges of the binding (facing) to the inside, insert the edges of sleeve opening and seam. The binding shall finish 1/2 to 3/4 inch (12.7 to 19.0 mm) wide.	301 or 401	BSc-1 BSc-1	10-14 10-14	70-2 70-2	70-2 80-2	
	b. Turn binding to inside of sleeve and stitch or bartack through binding and sleeve 1/8 to 1/4 inch (3.2 to 6.4 mm) below top of opening, the full-width of the binding.	301 or Bartack or 301		10-14 28 per bartack 28 per tack	70-2 70-2 70-2	70-2 70-2 70-2	
	- or -						
	c. Turn binding to inside of sleeve and stitch or bartack diagonally through binding only. An automatic straight tack may be used.	301 or Bartack or 301		10-14 28 per bartack 28 per tack	70-2 70-2 70-2	70-2 70-2 70-2	
13.	Join shoulder seam.						
	Position fronts between the two plies of yoke and with front and yoke edges even, stitch all plies together. The stitching shall not show on the finished shirt.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 80-2	

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TABLE II

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
14.	Set sleeves.						
	Set sleeves by overedging and seaming the raw edges together in one operation.	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 80-2	80-2 80-2 (over-edge)	
15.	Join side and underarm seams.						
	Join side and underarm seams by overedging and seaming the raw edges together in one operation, with the sleeve-setting seam turned toward the sleeve.	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 80-2 (over-edge)	80-2 80-2	
16.	Hem shirt.						
	Turn bottom of fronts and back of shirt with raw edges turned in, and stitch through shirt, with the hem extending across fold of fronts. Hem shall finish 1/8 to 3/16 inch (3.2 to 4.8 mm) wide. The side seams shall be turned in the sme direction as turned at the cuff.	301	EFb-1	10-14	70-2	70-2	
17.	Join cuffs to sleeves.						
	a. Seam the inside of cuff to bottom of sleeve on the underside, with the bound edge of the sleeve opening turned to the inside at buttonhole end of cuff and caught in the stitching.	301 or 401	LScf-2(a) LScf-2(a)	10-14 10-14	70-2 70-2	70-2 80-2	

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TABLE II

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
17.	Join cuffs to sleeves. (cont'd)						
	b. Stitch top edge of outer side of cuff, with interlining 1/16 to 1/8 inch (1.6 to 3.2 mm) from edge, to the bottom of sleeve on the outside.	301	LScf-2(b)	10-14	70-2	70-2	
	or						
	Cuffs may be attached in one operation, (see footnote 1/ to table I) providing the finished appearance is the same as specified in "a" and "b" above, and the ends of the row of stitching are backtacked or crossed with another line of stitching.	301	LSe-1	10-14	70-2	70-2	
18.	Join collar to shirt.						
	Finished appearance. The collar shall be carefully joined to the shirt, with notches matching, without distorting the stand or collar. Collar ends shall not be uneven in length by more than 1/8 inch (3.2 mm). In stitching down collarstand, the stitching shall be not more than 1/8 inch (3.2 mm) on or off the opposite stand.						
	a. Fold right and left fronts of shirt to the inside as indicated by marks on pattern. The facings shall finish not less than 1-1/4 (32 mm) or more than 1-3/4 inches (44 mm) wide.						
	b. Turn in the bottom edge of the topcollar stand and stitch 1/16 to 1/8 inch (1.6 to 3.2 mm) from turned edge.	301	LSb-1	10-14	70-2	70-2	
	- or -						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
18.	Join collar to shirt. (cont'd)						
	c. Stitch topcollar stand to inside of shirt, turn seam allowances toward stand.	301	LScf-2(a)	10-14	70-2	70-2	
	d. Stitch down the undercollar stand 1/16 to 1/8 inch (1.6 to 3.2 mm) from turned edge through stand and shirt; continue the stitching around the ends of the collarstand if not previously stitched (see operation 7.c).	301	LScf-2(b)	10-14	70-2	70-2	
19.	Make buttonholes in collarstand and front.						
	a. Mark or gage and make five vertical buttonholes in the left front of shirt through the facing as indicated by marks on pattern. They shall be 3-1/2 ± 1/4 inches (89 ± 6 mm) apart, measured from center to center.	304	whip or purl types	42-46 per buttonhole including tack	70-2	70-2	
	b. Make one horizontal buttonhole in center of collar stand with front edge of buttonhole extended 1/8 inch (3.2 mm) beyond line of front vertical buttonholes.	304	whip or purl types	43-46 per buttonhole including tack	70-2	70-2	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
20.	Sew on buttons. Finished appearance. The buttons shall be securely sewn and properly aligned with buttonholes in collarstand, front of shirt, and cuffs, Collar ends shall be centered equidistant from center of shirt front and shall open not less than 1/4 inch (6.4 mm) or more than 3/4 inch (19.0 mm) when collar is buttoned.						
	a. Sew five buttons on right shirt front 5/8 to 3/4 inch (15.9 to 19.0 mm) from front edge through button facing.	301 or		14-16 per button	70-2	70-2	
		101		14-16 per button	70-2		
	b. Sew one button on right end of collarstand so that when collar is buttoned, center of button shall be in line with vertical buttonholes on front without distorting left or right front.	301 or		14-16 per button	70-2	70-2	
		101		14-16 per button	70-2		
	c. Sew one button on the outside of each cuff at the back end, to correspond with the buttonhole, with the center of the button 3/8 to 5/8 inch (9.5 to 15.9 mm) from edge.	301 or		14-16 per button	70-2	70-2	
		101		14-16 per button	70-2		

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS TABLE II	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN or LOOPER	COVER
21.	Clean shirts. a. Trim all ends of thread and remove loose threads from shirts. Remove all spots and stains. b. Remove all shade tickets.						
22.	Pressing. a. As an option to operations 10.b and 11.b a hot head pressing machine capable of pressing the entire shirt after fabrication may be used (see 6.6). b. Press the bottom, side seams, underarm seams and crease sleeves from top of sleeve to bottom of cuff on a hot head pressing machine. c. When required touch up fronts, backs, cuffs and collars, before folding for packaging with a heated iron at a permanent press (moderate) setting. NOTE: The use of steam inflated bag type pressing machine is prohibited. The hot head pressing machine cited in operations 5.e, 7.d, 10.b, and 11.b shall be equipped with controls for accurate temperature (350 to 370°F), (177 to 188°C), time (minimum 10 seconds) and pressure (100 psi gauge) (689 kPA).						

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3.9 **Finished measurements.** The finished measurements shall conform to table III.

TABLE III. Finished measurements (inches and millimeters).

Size	Collar 1/	1/2 Chest 2	Back length 3/	Sleeve length 4/
13 (330)	13-1/2 (343)	19-1/4 (489)	31-1/4 (794)	1/2 inch (12.7 mm) over marked size (applicable to all sizes)
13-1/2 (343)	14 (356)	20-1/4 (514)	31-1/4 (794)	
14 (356)	14-1/2 (368)	21-1/4 (540)	31-1/2 (800)	
14-1/2 (368)	15 (381)	22-1/2 (572)	31-1/2 (800)	
15 (381)	15-1/2 (394)	23-1/2 (597)	31-1/2 (800)	
15-1/2 (394)	16 (406)	24-1/2 (622)	31-3/4 (806)	
16 (406)	16-1/2 (419)	25-1/2 (648)	31-3/4 (806)	
16-1/2 (419)	17 (432)	26-1/4 (667)	31-3/4 (806)	
17 (432)	17-1/2 (444)	27-1/4 (692)	32 (813)	
17-1/2 (444)	18 (457)	28-1/4 (718)	32 (813)	
18 (457)	18-1/2 (470)	29-1/4 (743)	32 (813)	
Tolerance:				
Plus	1/8 (3.2)	1/2 (12.7)	1/2 (12.7)	1/4 (6.4)
Minus	1/8 (3.2)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)

1/ Collar measurement taken from center of collar button to front cut-end of buttonhole, with collar in flat position.

2/ Chest measurement taken from folded edge to folded edge at base of armhole with shirt buttoned.

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3/ Back length measurement taken at center back from collar setting seam to bottom edge of shirt.

4/ Sleeve length measurement taken from center back at collar stand seam, diagonally across back to lower edge of the back yoke and down sleeve to bottom edge of cuff.

3.10 Workmanship. The finished shirts shall conform to the quality and grade of product established by this specification. As a final step in the supplier's production control plan before formation of a lot, each shirt shall be examined, buttoned, pressed and laid out flat on a table prior to folding for packaging. A shirt containing a selected defect shall not be included in the end item lot. Selected defects are those defects listed in MIL-STD-1492, indicated by an asterisk (*). The occurrence of defects shall not exceed the applicable point value or defect limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1492, except where otherwise indicated hereinafter.

4.2.1 Component and material testing. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on the components listed in table IV for characteristics noted. The method of testing specified in FED-STD-191, wherever applicable, and so listed in table IV shall be followed. All requirements are applicable to the sample unit. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

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The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The unit for expressing lot size, and the sample unit for testing each component shall be in accordance with applicable subsidiary specifications and as follows:

Component	Lot size expressed as	Sample unit for testing
Fusible interlining for collar leaf and cuffs	Yards	2 yard full width
Fusible interlining fused to basic cloth	Yards	1 yard full width
Collar stays	Gross	4 each

TABLE IV. Test methods

Component	Characteristics	Requirement paragraph	Test method
Fusible interlining for collar leaf and cuffs	Weave	3.2.2.2	Visual
	Weight	3.2.2.2	5041
	Thread per inch:	3.2.2.2	5050
	-warp		
	-filling		
	Shrinkage	3.2.2.2	5556.1
	-warp		
	-filling		
Fusible interlining fused to basic cloth	Stiffness	3.2.2.2	5202
Fusible interlining fused to basic cloth (after 25 launderings)	Adhesion	3.2.2.2	4.3.1.3 and 5951
Collar stays	Width	3.2.6	Gage
	Length	3.2.6	Gage

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

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4.2.3 End item inspection. The shirts shall be tested as specified in table V. All test reports shall contain the individual values utilized in expressing the final results. All requirements shall be applicable to the sample unit. The sample size shall be as specified in 4.2.1 and the sample unit shall be one finished shirt. One shirt shall be used to evaluate the appearance requirements. The lot shall be unacceptable if one or more units fail to meet the requirement specified.

TABLE V. End item testing

Characteristic	Requirement paragraph	Test method
Seam appearance rating	3.6	4.3.1
Bubbling or delamination of collars or cuffs	3.6	4.3.1

4.2.4 Quality conformance inspection. Sampling and examination of the end item shall be performed in accordance with MIL-STD-1492. The shirt shall be classified as a dress ("D") shirt.

4.3 Test methods.

4.3.1 Appearance of seams, bubbling or delamination of collar and cuffs and adhesion of fusible interlining to basic cloth.

4.3.1.1 Apparatus and material used for laundering. The following apparatus and material is required for laundering:

a. Automatic washing machine (see 6.6).
b. Automatic dryer (see 6.6).
c. AATCC Standard Detergent 124 or equivalent.

4.3.1.2 Apparatus and material used for evaluation. The following apparatus and material is required for evaluation of samples for seam appearance:

Overhead lighting procedure. Apparatus shall be as specified in AATCC Test Method 124-1969. In addition AATCC photographic standards for evaluating single needle stitching are required. Evaluation shall be conducted in a room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black, or blackout curtains may be hung on each side of the viewing area.

4.3.1.3 Procedure.

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4.3.1.3.1 Laundering and conditioning. If the shirts selected as test samples weigh less than 4 pounds (1.8 kg), ballast to make up a standard 4 pound (1.8 kg) load shall be added to the test shirts. The ballast shall consist of durable press shirts that are approximately the same weight as the test shirts or pieces of durable press fabric that are similar to the fabric of the test shirts. The pieces shall be hemmed and measure 3 feet by 3 feet (0.91 m by 0.91 m). The front of each shirt in the 4 pound (1.8 kg) load shall be unbuttoned. The load shall be placed in the washing machine. Add 90 grams of detergent to the washing machine. The washing machine shall be set for 12 minutes on the hot wash and wear cycle, high water level. The washing machine shall be filled with water from a water source set at 140° ± 5°F (60° ± 3°C). Immediately after the washing machine has completed the final spin cycle, the entire load shall be removed, untangled, and placed in the dryer. The dryer shall be operated at a temperature of 140° to 160° (60° to 71°C) until the entire load is dry, continue tumbling 5 minutes with the heat turned off (cool-down cycle). Remove the load immediately after the machine stops. Repeat the wash, dry and tumble cycles until the load has completed 5 complete cycles. After completion of the fifth cycle, the test shirts shall be conditioned for 2 hours at 70° ± 10°F (21° ± 6°C) and at a relative humidity of 50 to 65 percent. (NOTE: During the interruptions between the cycles and during the conditioning and the evaluations, the test shirts shall be placed on wooden or plastic hangers and shall be buttoned at the front).

4.3.1.4 Evaluation of the appearance of seams. After the test shirts have been laundered and conditioned, the shirts shall be evaluated by three trained observers. The trained observers, independent of the other observers, shall evaluate each of the shirts as follows: The shirts shall be placed on a clothes hanger with the buttons fastened and the hanger suspended on the viewing board in such a manner that the shirt lies flat on the surface of the board. The photographic standards for single needle stitching shall be placed on the viewing board. The observer shall stand directly in front but 4 feet (1.22 m) from the test shirt. The observer shall then compare the test shirt, in each of the areas specified in 3.6, to the photographic standards and shall assign a rating value to each of the specified areas. The rating value assigned to an area by the observer shall be the numerical value of the photographic standard that most nearly matches the area. No estimation of ratings falling between standards are allowed. Each rating value assigned by each of the observers shall be recorded. The average of the rating values assigned by each observer to the same area shall be the rating value for that area. After the shirts have been rated for seam appearance (5 launderings) they shall be laundered an additional 20 times as specified in 4.3.1.3.1 (25 total launderings) and evaluated visually for bubbling or delamination of the collars and cuffs. In addition, during the laundering of the shirts, 1/2 yard full width of the fusible interlining fused to the basic cloth shall also be laundered 25 times. After completion of the 25 launderings, the fused cloth shall be tested for adhesion as specified in 4.2.1.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each fully buttoned shirt shall be neatly folded to measure approximately 14-1/2 inches by 10-1/2 inches (368 mm by 267 mm). The folded shirt shall be secured with stainless steel (AISI type 304 or 316) pins, aluminum clips, or plastic fasteners. Each shirt shall be inserted in a snug-fitting flat style clear polyethylene film bag of 0.00125 inch (0.0317 mm) thickness (± 25 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight and continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. A 1/4 inch (6 mm) diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Level C. Shirts shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Seventy-two shirts of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, grade V2s, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A packages shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2 Level B. Seventy-two shirts of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 200, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A packages shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth with the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

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5.2.3 Level C. Shirts, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transporation rate for such supplies. Containers shall be in accordance with the U. S. Postal Service Manual, Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethyelene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches (127 by 102 mm) with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The shirts are intended for wear by male military personnel of the Department of the Army.

6.2 Ordering data. Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples address the procuring office issuing the invitation for bids.

6.4 Metric units. Metric equivalents for customary inch units are shown in this specification. This information is provided only to familiarize readers with units that are expected ultimately to be adopted nationally.

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6.5 In order to avoid puckering and to produce the flattest seams possible, consideration must be given to the following:

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate hole should be no larger than necessary to permit free entry of the needle and to prevent flagging.

c. The finest feed (approximately 18 teeth) possible should be used with a pressure foot to exactly match the shape and size of the feed. The face of the pressure foot (underside next to the feed on the machine) should be highly polished to eliminate roughness and reduce drag. Diamond shaped feed teeth have been used quite successfully.

d. The lightest pressure foot pressure should be used to prevent stretching of the top ply of material.

e. In the safety stitch machine (515, 516 and 519), the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the edge of the fabric.

f. In stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

g. When sewing with stitch type 401 (including the seaming row of the safety stitches), the thread ratio (needle and looper) should be adjusted to produce a stitch with 60 to 70 percent needle thread, and both needle and looper thread tension should be held to the minimum.

h. Thread takeup mechansim should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free of burrs, cuts and rough spots.

6.6 The recommended apparatus for 4.3.1.1 a and b., and for pressing in operation 22. a are:

a. Kenmore Automatic Washing Machine Model 600, or similar washing machine. In case of dispute, the Kenmore Model 600 shall be used.

b. Kenmore Automatic Dryer Model 600, or equivalent.

c. "Formatic" hot head pressing machine or equal. "Formatic" manufactured by the American Laundry Machine Co., Cincinnati, OH.

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6.7 Fusing conditions. The following fusing conditions have been found to satisfactorily obtain the required results of this document. These limits are furnished for guidance only:

	Minimum	Maximum
Temperature (between pressing surfaces), °F	320	370
Pressure (between pressing surfaces), PSI	20	40
Pressing time, seconds	14	20

It is recommended that adhesion tests on the collar and cuffs be performed with a spring scale at the fusing press, twice a day, to assure that adequate fusing is being attained. An adhesion of 30 pounds per inch has been found to be sufficient.

6.8 Supersession data. This specification includes the requirements in part of the type I, class 2 shirt specified in MIL-S-41827F dated 6 September 1973.

Custodian:

Army - GL

Review activities:

Army - MD
DSA - CT

Preparing activity:

Army - GL

Project No. 8405-A672

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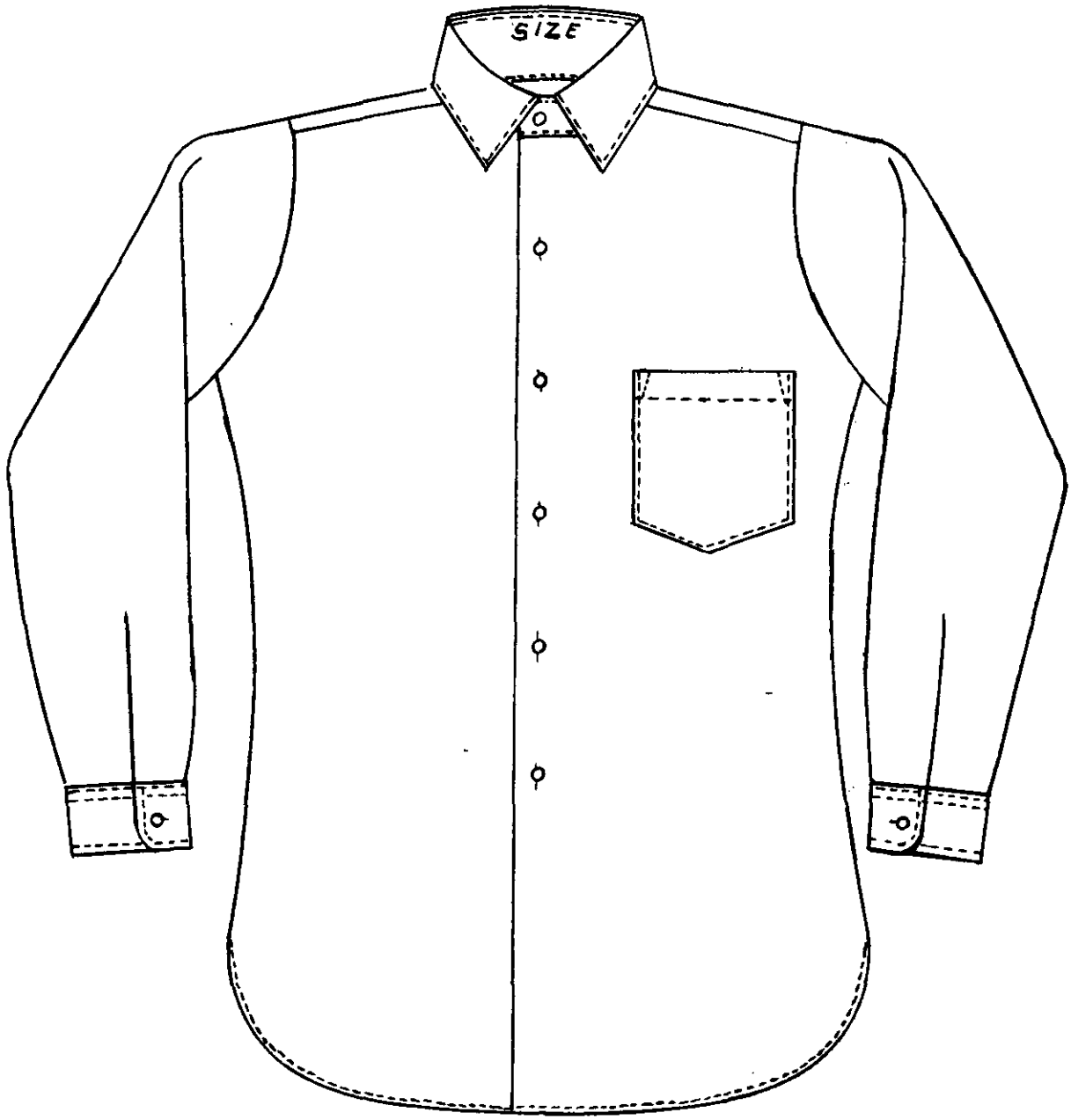


FIG. 1 SHIRT, MAN'S, POLYESTER/COTTON DURABLE PRESS

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-43939(GL) SHIRT, MAN'S, POLYESTER/COTTON, DURABLE PRESS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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